

WEEK 5 – PAYMENT TRANSACTION MANAGEMENT SYSTEM

CITYCARE HOSPITAL MANAGEMENT SYSTEM

1. Introduction

The Payment Transaction Management System is an important module of the CityCare Hospital Management System. It is used to maintain payment information related to patient orders, record transaction details, track payment status, and generate useful payment and revenue reports.

The system stores details such as Payment ID, Order ID, Payment Date, Payment Mode, Payment Status, and Transaction Amount.

2. Objective

The main objectives of the Payment Transaction Management System are:

- To create a payment table for storing transaction details.
- To record payment information for patient orders.
- To track successful, failed, and pending payments.
- To identify failed transactions.
- To analyse different payment modes.
- To calculate total revenue and average transaction amount.
- To generate patient payment history reports.

These objectives correspond to the Week 5 payment-management requirements in the reference material.

3. Payment Table Structure

The Hospital_Payment table is created to store payment transaction details.

Attributes

Attribute	Description
Payment_ID	Unique identification number for payment
Order_ID	Identifies the related patient order
Payment_Date	Date of payment
Payment_Mode	Mode used for payment
Payment_Status	Status of the transaction
Transaction_Amount	Amount involved in the transaction

The payment modes used are UPI, Card, Net Banking, and Cash. The payment statuses are Success, Failed, and Pending.

4. Table Creation

The following SQL command creates the Hospital Payment table:

```
SQL> CREATE TABLE Hospital_Payment (  
2     Payment_ID NUMBER PRIMARY KEY,  
3     Order_ID NUMBER NOT NULL,  
4     Payment_Date DATE DEFAULT SYSDATE,  
5     Payment_Mode VARCHAR2(20) NOT NULL,  
6     Payment_Status VARCHAR2(20) DEFAULT 'Pending',  
7     Transaction_Amount NUMBER(10,2) CHECK (Transaction_Amount > 0),  
8     CONSTRAINT fk_payment_order FOREIGN KEY (Order_ID)  
9         REFERENCES Patient_Orders(Order_ID),  
10    CONSTRAINT chk_payment_mode CHECK  
11        (Payment_Mode IN ('UPI','Card','Net Banking','Cash')),  
12    CONSTRAINT chk_payment_status CHECK  
13        (Payment_Status IN ('Success','Failed','Pending'))  
14 );
```

Table created.

```
SQL> DESC Hospital_Payment;
```

Name	Null?	Type
PAYMENT_ID	NOT NULL	NUMBER
ORDER_ID	NOT NULL	NUMBER
PAYMENT_DATE		DATE
PAYMENT_MODE	NOT NULL	VARCHAR2(20)
PAYMENT_STATUS		VARCHAR2(20)
TRANSACTION_AMOUNT		NUMBER(10,2)

5. Inserting Payment Records

Payment details are inserted into the Hospital_Payment table for the existing patient orders.

```
SQL> INSERT INTO Hospital_Payment
  2 (Payment_ID, Order_ID, Payment_Mode, Payment_Status, Transaction_Amount)
  3 VALUES
  4 (701, 501, 'UPI', 'Success', 13.00);

1 row created.

SQL>
SQL> INSERT INTO Hospital_Payment
  2 (Payment_ID, Order_ID, Payment_Mode, Payment_Status, Transaction_Amount)
  3 VALUES
  4 (702, 502, 'Card', 'Success', 17.50);

1 row created.

SQL>
SQL> INSERT INTO Hospital_Payment
  2 (Payment_ID, Order_ID, Payment_Mode, Payment_Status, Transaction_Amount)
  3 VALUES
  4 (703, 503, 'Cash', 'Success', 10.00);

1 row created.

SQL>
SQL> INSERT INTO Hospital_Payment
  2 (Payment_ID, Order_ID, Payment_Mode, Payment_Status, Transaction_Amount)
  3 VALUES
  4 (704, 504, 'Net Banking', 'Failed', 12.00);

1 row created.

SQL>
SQL> COMMIT;

Commit complete.
```

6. Display Successful Payments

Successful transactions can be displayed using the following query:

```
SQL> SELECT * FROM Hospital_Payment
2 ORDER BY Payment_ID;
```

PAYMENT_ID	ORDER_ID	PAYMENT_D	PAYMENT_MODE	PAYMENT_STATUS
701	501	05-OCT-26	UPI	Success
702	502	05-OCT-26	Card	Success
703	503	05-OCT-26	Cash	Success
704	504	05-OCT-26	Net Banking	Failed

7. Find Failed Transactions

Failed payment transactions can be identified using:

```
SQL> SELECT *
2 FROM Hospital_Payment
3 WHERE Payment_Status = 'Success';
```

PAYMENT_ID	ORDER_ID	PAYMENT_D	PAYMENT_MODE	PAYMENT_STATUS
701	501	05-OCT-26	UPI	Success
702	502	05-OCT-26	Card	Success
703	503	05-OCT-26	Cash	Success

8. Count Successful and Failed Payments

The number of transactions for each payment status can be calculated using:

```
SQL> SELECT
  2     Payment_Status,
  3     COUNT(*) AS Total_Transactions
  4 FROM Hospital_Payment
  5 GROUP BY Payment_Status;
```

PAYMENT_STATUS	TOTAL_TRANSACTIONS
Failed	1
Success	3

9. Payment Mode Analysis

Payment Mode Analysis is used to determine how many transactions were made using each payment method. The Week 5 reference specifically includes analysing UPI, Card, and the most preferred payment method.

Payment Mode-wise Transaction Count

```
SQL> SELECT
  2     Payment_Mode,
  3     COUNT(*) AS Number_of_Transactions
  4 FROM Hospital_Payment
  5 GROUP BY Payment_Mode
  6 ORDER BY Number_of_Transactions DESC;
```

PAYMENT_MODE	NUMBER_OF_TRANSACTIONS
Net Banking	1
UPI	1
Cash	1
Card	1

Most Preferred Payment Method

```
SQL> SELECT
  2     Payment_Mode,
  3     COUNT(*) AS Transaction_Count
  4 FROM Hospital_Payment
  5 GROUP BY Payment_Mode
  6 HAVING COUNT(*) = (
  7     SELECT MAX(COUNT(*))
  8     FROM Hospital_Payment
  9     GROUP BY Payment_Mode
 10 );
```

PAYMENT_MODE	TRANSACTION_COUNT
Net Banking	1
Card	1
Cash	1
UPI	1

This query identifies the payment method with the highest number of transactions.

10. Revenue Analysis

Revenue Analysis is used to calculate the revenue generated through successful payment transactions. The Week 5 reference includes total revenue, revenue by payment method, and average transaction amount.

10.1 Total Revenue

```
SQL> SELECT
  2     SUM(Transaction_Amount) AS Total_Revenue
  3 FROM Hospital_Payment
  4 WHERE Payment_Status = 'Success';
```

TOTAL_REVENUE
40.5

This query calculates the total amount received from successful transactions.

10.2 Revenue by Payment Method

```
SQL> SELECT
  2     Payment_Mode,
  3     SUM(Transaction_Amount) AS Revenue
  4 FROM Hospital_Payment
  5 WHERE Payment_Status = 'Success'
  6 GROUP BY Payment_Mode
  7 ORDER BY Revenue DESC;
```

PAYMENT_MODE	REVENUE
Card	17.5
UPI	13
Cash	10

This query calculates the revenue generated through each successful payment method.

10.3 Average Transaction Amount

```
SQL> SELECT
  2     ROUND(AVG(Transaction_Amount), 2)
  3     AS Average_Transaction_Amount
  4 FROM Hospital_Payment
  5 WHERE Payment_Status = 'Success';
```

AVERAGE_TRANSACTION_AMOUNT
13.5

This query calculates the average amount of successful payment transactions.

11. Patient Payment History

The Patient Payment History report displays the patient name, order ID, payment mode, transaction amount, and payment status.

```
19.8
SQL> SELECT
  2     p.Patient_Name,
  3     po.Order_ID,
  4     hp.Payment_Mode,
  5     hp.Transaction_Amount,
  6     hp.Payment_Status
  7 FROM Patient p
  8 JOIN Patient_Orders po
  9     ON p.Patient_ID = po.Patient_ID
 10 JOIN Hospital_Payment hp
 11     ON po.Order_ID = hp.Order_ID
 12 ORDER BY po.Order_ID;
```

PATIENT_NAME	ORDER_ID	PAYMENT_MODE	TRANSACTION_AMOUNT	PAYMENT_STATUS
Ananya Kumar	501	UPI	13	Success
Rahul Sharma	502	Card	17.5	Success
Meena Devi	503	Cash	10	Success

PATIENT_NAME	ORDER_ID	PAYMENT_MODE	TRANSACTION_AMOUNT	PAYMENT_STATUS
Arun Raj	504	Net Banking	12	Failed

This report connects the Patient, Patient_Orders, and Hospital_Payment tables to provide complete payment information for each patient.

12. Final Payment Report

A complete payment report can be generated using the following SQL query:

```
SQL> SELECT
2     p.Patient_Name,
3     po.Order_ID,
4     hp.Payment_Date,
5     hp.Payment_Mode,
6     hp.Payment_Status,
7     hp.Transaction_Amount
8 FROM Patient p
9 JOIN Patient_Orders po
10    ON p.Patient_ID = po.Patient_ID
11 JOIN Hospital_Payment hp
12    ON po.Order_ID = hp.Order_ID
13 ORDER BY po.Order_ID;
```

PATIENT_NAME	ORDER_ID	PAYMENT_D	PAYMENT_MODE	PAYMENT_STATUS
Ananya Kumar	501	05-OCT-26	UPI	Success
	13			
Rahul Sharma	502	05-OCT-26	Card	Success
	17.5			

PATIENT_NAME	ORDER_ID	PAYMENT_D	PAYMENT_MODE	PAYMENT_STATUS
Meena Devi	503	05-OCT-26	Cash	Success
	10			
Arun Raj	504	05-OCT-26	Net Banking	Failed

```
17.5
```

PATIENT_NAME	ORDER_ID	PAYMENT_D	PAYMENT_MODE	PAYMENT_STATUS
Meena Devi	503	05-OCT-26	Cash	Success
	10			
Arun Raj	504	05-OCT-26	Net Banking	Failed

PATIENT_NAME	ORDER_ID	PAYMENT_D	PAYMENT_MODE	PAYMENT_STATUS

12

13. Conclusion

The Payment Transaction Management System successfully manages payment information in the CityCare Hospital Management System. It records payment transactions, identifies successful and failed payments, analyses payment methods, calculates revenue, and provides patient payment history.

Thus, the Week 5 module helps maintain organized payment records and provides useful reports for analysing hospital transactions.